

BANARAS HINDU UNIVERSITY
B.A. (Hons.) Semester II Examination, 2022-23

Paper No. STA-201: Descriptive Statistics and Distribution Theory

Time: 3 Hours

Maximum Marks: 70

*Instructions: Answer **five** questions including Question 1 which is compulsory. Symbols have their usual meanings.*

Question 1.

- (a) Define mathematical expectation. If X is a random variable, then prove that $|E(X)| \leq E(|X|)$. [#1]
- (b) Write a note on linear and non-linear correlation. Also show that Karl Pearson's coefficient of correlation lies between -1 and +1. [#1]
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Question 2.

- (a) If X_1 , X_2 , and X_3 are uncorrelated variables each having the same variance, obtain the correlation coefficient between $X_1 + X_2$ and $X_2 + X_3$. [#1]
- (b) A random variable X can assume any positive integral value n with a probability proportional to $1/3^n$. Find the expectation of X . [#1]
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Question 3.

- (a) Define intra-class correlation and derive the expression for the intra-class correlation coefficient. [#1]
- (b) Define convergence in probability. Let $X_1, X_2, \dots, X_n$ be i.i.d. random variables with probability density function $f(x) = e^{-x}, x > 0$. Show that $Y_n \rightarrow 0$ in probability, where $Y_n = \min(X_1, X_2, \dots, X_n)$. [#1]
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Question 4.

- (a) For the joint distribution of two-dimensional random variables (X, Y) given by:

$$f(x, y) = e^{-(x+y)}, \quad x > 0, y > 0$$

Show that the characteristic function of $X + Y$ is equal to the product of the characteristic functions of X and Y . [#1]

- (b) Find the value of the correlation coefficient for the data set (X, Y) in which Y has the same constant value corresponding to any value of X . [#1]
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Question 5.

- (a) For two variables X and Y with the same mean, the two regression equations are $y = ax + b$ and $x = \alpha y + \beta$. Find the mean and show that $\frac{b}{\beta} = \frac{1-a}{1-\alpha}$. [#1]
- (b) Define the consistency of data. If data has three attributes, give the conditions of the consistency of the data. [#1]
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Question 6.

- (a) Derive the Poisson distribution as a limiting case of the Binomial distribution. Also obtain the mode of the Poisson distribution. [#1]
- (b) Write a note on the lack of memory property of the geometric distribution. [#1]
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Question 7.

- (a) Define the gamma distribution with one as well as two parameters. Derive the moment generating function for the two-parameter gamma distribution. [#1]
- (b) If X is uniformly distributed with mean 1 and variance $4/3$, find $P(X < 0)$. [#1]
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Question 8. Write a note on any **three** of the following: [#1]

- (a) Rank Correlation
- (b) Normal Distribution
- (c) Association of Attributes
- (d) Chebyshev's Inequality
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